

AHMED MOHAMED AMIN

+20 128 802 4972 | Giza, Egypt | s-ahmed.mohamed@zewailcity.edu.eg

[LinkedIn](#) | [GitHub](#) | [Portfolio](#) | [IEEE Publication](#)

EDUCATION

Zewail City of Science and Technology

Giza, Egypt

B.Sc. Communication & Information Engineering (ABET) | GPA: 3.63/4.0 (Cum Laude) Sept 2021 – June 2026

- **Key Courses:** Machine Learning, Deep Learning, NLP, Computer Vision, Algorithms & Data Structures, Software Engineering, Database Systems, Computer Architecture, Probability & Statistics, Linear Algebra, Signal Processing, Operating Systems, Computer Networks.
- **Honors:** Golden Star Award (GPA > 3.5) for **7 consecutive semesters**; 5-year full Excellence Scholarship.

PUBLICATION

IEEE CSCloud 2025 — 12th International Conference on Cyber Security and Cloud Computing

Aug 2025

- L. Rana, **A. M. Amin**, H. Ameer, and S. Latif, “FinVerse Agent: Resource-Efficient Financial Language Models with Modular LoRA Fine-Tuning and Task Routing,” in *Proc. IEEE CSCloud*, 2025.
- **My contribution:** Built the modular LoRA adapter framework and the multi-task fine-tuning pipeline across 15 financial NLP tasks, cutting compute cost by **10×** compared to full fine-tuning.

RESEARCH EXPERIENCE

National University of Sciences and Technology (NUST)

Islamabad, Pakistan

AI Research Intern — SEECS Lab

June 2025 – Aug 2025

- Matched BloombergGPT performance while using **75% less memory**: reached **90.1%** accuracy on CFA exam questions, **89.6% F1** on financial named-entity recognition, and **85.1%** on FIQA sentiment, with less than 1% accuracy loss from 4-bit quantization.
- Won **1st Place (Task 2)** at FinAI 2025 (IEEE CSCloud) by fine-tuning Llama 3.1 8B and Qwen 2.5 with **LoRA and QLoRA** on CFA, XBRL, and Bloomberg datasets.
- Designed a modular fine-tuning pipeline in Unsloth with **15 task-specific LoRA adapters** for classification, numerical reasoning, and NER on OCR-processed financial documents.

INDUSTRY EXPERIENCE

Takhail | AI Engineer | Giza, Egypt

Dec 2025 – Present

- Built a production **multilingual RAG pipeline** (ChromaDB, FastAPI, Celery, Redis) on AWS EC2 with Docker Swarm for a luxury real estate assistant supporting Arabic and English; implemented hybrid BM25 + vector retrieval with semantic query routing based on intent, language, and topic.
- Designed an **async fire-and-poll architecture** with Celery task queues, cutting response blocking and enabling parallel STT, TTS, and RAG processing through a single unified voice and chat API.
- Built a **RAG observability pipeline** using structured stdout traces shipped via Promtail to Loki, with a token-gated live debug dashboard showing real-time pipeline stages, retrieval scores, and LLM timing — keeping the user database clean of debug data.
- Designed the full software architecture including API design, containerized microservices (MinIO, ChromaDB, MongoDB, Redis), ElevenLabs TTS, Whisper STT, and multi-provider LLM switching (OpenAI, OpenRouter/DeepSeek) with zero-downtime Swarm deployments.

Rafeeq — Smart Wheelchair AI Companion | *Lead AI Engineer* | *Arabic NLP, Edge AI, IoT* Sept 2025 – 2026

- **AI & Edge:** Built offline Egyptian Arabic speech recognition on Raspberry Pi using MARBERT, TFLite, and EfficientNet-Lite. Added binary facial emotion detection and real-time health monitoring (heart rate, SpO₂, temperature) with fall detection and emergency calls through Twilio. Designed the system to be privacy-first, with no cloud image storage.
- **Software:** Built a 22-screen Flutter mobile app (Riverpod, 43 packages) with a RAG-powered Egyptian Arabic chatbot backend. Built the sensor-fusion architecture on AWS and Docker with FastAPI microservices.
- **Funding:** Won a **\$1,600 USD grant** from Egypt's Ministry of Higher Education through the "Your Project, Your Startup" national program, selected among projects from all Egyptian universities (Mar. 2026).

SELECTED PROJECTS

Arabic Medical Consultation Chatbot | *RAG, Arabic NLP, FastAPI, Docker, MongoDB* 2025

- Built a bilingual Arabic and English medical assistant using retrieval-augmented generation with TTS voice output. Backend uses FastAPI, Celery, MongoDB, and Redis, with persistent conversation history.

Medical Semantic Search Engine | *ELMo, BiLSTM, FAISS, FastAPI* Mar – May 2025

- Built a CPT code classifier that reached **87% Top-3 accuracy** with sub-second retrieval. Used contextual ELMo embeddings with a BiLSTM for semantic search over 10K+ clinical notes, served through FastAPI.

DSTN for Click-Through Rate Prediction | *PyTorch, ConvLSTM, Conv3D* Aug – Sept 2025

- Implemented the dual-stream spatio-temporal network from the KDD 2019 paper, using ConvLSTM and Conv3D to model spatial ad context and temporal user behavior together on large-scale advertising data.

TECHNICAL SKILLS

Languages: Python, C++, SQL

NLP & LLMs: LoRA/QLoRA, 4-bit Quantization, RAG, LangChain, Hugging Face Transformers, Unsloth, vLLM, Whisper, TTS

ML & Vision: PyTorch, TensorFlow/TFLite, Scikit-learn, YOLO, RF-DETR, CLIP, 3D-CNN, OpenCV, EfficientNet, PEFT

Infrastructure: AWS EC2, Docker, Docker Swarm, Traefik, FastAPI, Redis, Celery, ChromaDB, FAISS, Git, Linux, CI/CD

Software Eng.: REST API Design, Microservices Architecture, System Design, Flutter/Riverpod, Agile/Scrum, MongoDB

LEADERSHIP & EXTRACURRICULARS

Egypt Scholars — Basic Lab 12 | *Competitively Selected Participant* Feb 2026 – May

- Selected for a 3-month competitive program on research methods, scientific writing, and critical thinking.

Elected Student Representative | *Zewail City of Science and Technology* 2021 – 2025

- Represented all CIE cohorts (4 batches) for 4 years in a row, working with faculty on academic matters that affected **200+ students**.

Junior Teaching Assistant — Math 101 | *Zewail City* Oct 2022 – Jan 2023

- Led tutorials and gave one-on-one mentoring to 50+ first-year students in calculus and linear algebra.